

Nova School of Business and Economics
Masters Programme in Business Analytics

2758-T4 Advanced Topics in Machine Learning

Final Group Project Report

VinhaGuard

*Parametric Climate Insurance Infrastructure
for Portugal's Douro Valley*

Course: 2758-T4 Advanced Topics in Machine Learning
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Programme: Masters in Business Analytics
Submission: May 2026

This document is submitted as the final written report for the AI Startup Brainstorm project.
All quantitative claims are traceable to the project repository. VinhaGuard is an academic
prototype and is not a licensed insurance product.

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Abstract

VinhaGuard is a vineyard-specific parametric climate insurance infrastructure layer for insurers and cooperatives serving Portugal's Douro Valley. The system uses thirty years of daily ERA5 reanalysis weather data from 32 Douro vineyard sites, enriched with TerraClimate monthly water-balance indicators, to classify climate-stress years, calibrate payout trigger thresholds, and produce transparent premium quotes. The deployed prototype achieves a ROC-AUC of 0.977 (calibrated Random Forest) on a chronological 2020–2024 holdout and correctly identifies all five documented extreme vintages (2003, 2005, 2017, 2022, 2024). The B2B2C business model positions VinhaGuard as a data, pricing, and monitoring layer partnered with a licensed insurer or cooperative, earning commission and platform-fee revenue without carrying underwriting risk. A Streamlit application demonstrates the full product workflow: risk assessment, pricing explainer, historical dashboard, and an AI assistant constrained to Douro climate-risk and parametric insurance topics. Generative AI tools were used extensively throughout the project for ideation, code scaffolding, documentation, and consistency checking; all use is disclosed in the GenAI transparency log (Section 10).

1 Introduction and Motivation

Small and medium wine producers in the Douro Valley face a combination of growing climate exposure and limited financial resilience. The region covers nearly 45,000 hectares of vineyards—approximately 22% of Portugal's total vineyard area—and accounts for over 40% of Portuguese wine exports santos2020. Traditional agricultural insurance is widely available in Portugal, but it is poorly suited to smallholders: field-assessment indemnity claims take weeks to settle, premium structures are designed for larger commercial operations, and post-event liquidity pressure is felt most acutely in the weeks immediately after a damaging climate event.

VinhaGuard addresses this gap through parametric insurance logic: instead of verifying losses with a field adjuster, the system pays when an objective, pre-agreed weather threshold is breached. Payout is automatic, auditable, and fast—the target is under 72 hours after trigger confirmation. The product is deliberately framed as *fast climate-liquidity protection*, not full crop-loss compensation.

The project was developed for the Advanced Topics in Machine Learning final group project at Nova SBE. It applies several ML concepts from the course (supervised classification, probability calibration, feature engineering, cross-validation design) to a real business problem with a deployed prototype.

1.1 Relevance of Machine Learning

The core business problem cannot be solved with a simple actuarial table. Three aspects require machine learning:

1. **Location-specific trigger calibration.** A single regional threshold applied uniformly across all Douro sites creates systematic basis risk. The risk model uses per-location 80th-percentile thresholds so triggers reflect local anomaly rather than absolute temperature.
2. **Multi-hazard risk scoring.** Heat, frost, and drought interact. A gradient over

elevation, longitude, and growing-season water balance determines which hazard dominates at each vineyard. Feature engineering and a Random Forest capture these non-linear interactions.

3. **Calibrated probability for pricing.** The premium formula requires a credible estimate of trigger probability. Isotonic calibration of the Random Forest improves the Brier score from 0.115 to 0.042, making risk scores suitable as pricing inputs.

2 Ideation Process

The team explored multiple startup concepts before settling on VinhaGuard. This section documents the ideation journey, as required by the project brief.

2.1 Initial Concepts Considered

Two alternative ideas were brainstormed and evaluated in detail before being set aside.

GovBid AI — The RFP Auto-Responder for SMBs

Problem identified. Local and federal governments release thousands of lucrative procurement contracts (RFPs) daily. Small-to-medium businesses miss out because reading a 100-page government RFP and formatting a compliant proposal takes weeks of human labour.

AI engine proposed. A three-agent workflow: (1) an *Extractor* agent that parses PDF RFPs and pulls exact compliance requirements; (2) a *RAG Matcher* agent that searches the company's vector database of past work, employee resumes, and policies; (3) a *Writer* agent that drafts the proposal section-by-section. A human-in-the-loop dashboard allows review and approval before submission.

Unit economics. Processing a 100k-token RFP and generating a 20k-token response with a model such as Gemini 1.5 Pro costs approximately \$1.50 per bid. The proposed revenue model was either a \$500/month SaaS subscription or a pay-per-bid fee of \$20 per generated bid—a significant ROI compared to a \$100/hour human grant writer.

Full-stack prototype stack. Streamlit for document upload and chat interface, LangChain/LlamaIndex for orchestration, Pinecone for the Vector DB, and Firebase for user authentication.

Why we moved on. While the concept is technically sound and the market is real, the team identified that government contracting knowledge in Portugal is highly fragmented and localised. Assembling a reliable corpus of past proposals and Portuguese procurement rules in the time available for the project was not feasible. The defensibility argument also relied on proprietary bid history, which the team did not have.

VendorBot — Autonomous B2B Negotiator

Problem identified. Small manufacturing and retail businesses lose margin because they lack the time and analytical tools to negotiate routine supply contracts. Procurement staff manage dozens of vendor relationships manually, with no data-driven benchmarking.

Solution proposed. An AI system that analyses historical invoices, integrates with company email, and drafts negotiation emails to vendors based on market pricing data.

The system would learn which suppliers are most price-flexible, which categories have the most margin room, and when to escalate a negotiation.

Moat argument. High switching costs once connected to the company’s ERP and email. Large horizontal AI providers would not build the deep ERP integration required for specific SMB accounting systems.

Why we moved on. The business analytics component was weaker than the other concepts: the value proposition was primarily workflow automation rather than predictive modelling. Access to invoice data and vendor price benchmarks for a prototype was also non-trivial within the project timeline.

2.2 Why We Chose VinhaGuard

After evaluating the alternatives, the team converged on VinhaGuard for the following reasons:

- **Business analytics centrality.** The product is fundamentally a data and risk-modelling product. The ML model, feature engineering, and pricing backend are the core value, not a wrapper around a general LLM.
- **Publicly available, high-quality data.** ERA5 reanalysis data via the Open-Meteo API and TerraClimate water-balance data are freely accessible and well-documented, enabling a fully reproducible pipeline without proprietary data access.
- **Local relevance.** The Douro Valley is a globally significant wine region and is immediately understandable to Portuguese and international audiences alike. The climate-risk narrative is timely given recent extreme vintages (2022, 2024).
- **Genuine ML problem.** Location-specific trigger calibration, multi-hazard classification, and probability calibration for pricing are substantive ML tasks well-suited to the course material.
- **Full prototype deployability.** The Streamlit stack and scikit-learn backend can be deployed end-to-end in the project timeline.

2.3 GenAI Role in Ideation

The team used Claude (Anthropic) and ChatGPT (OpenAI) to generate and stress-test the initial list of startup concepts. GenAI was asked to propose AI-native business models with strong data and analytics components, then to critique each idea on feasibility, data availability, and moat defensibility. The GovBid AI and VendorBot concepts above emerged from this structured brainstorm. The decision to select VinhaGuard was made by the team after reviewing GenAI assessments of all three options against the project evaluation criteria.

3 Business Problem and Market

3.1 The Pain Point

The VinhaGuard dataset validates the problem empirically. Across 32 vineyard sites and 30 years (1995–2024), **39.8% of all location-years qualify as climate stress years** under the model’s composite label. Table 1 illustrates the recurrence and severity of documented extreme vintages.

Table 1: Documented extreme vintages in the Douro Valley (1995–2024)

Year	% of 32 sites classified as stress	Known event
2003	81%	European heatwave — hottest summer in 500 years
2005	94%	Severe Iberian drought
2017	94%	Record heat and Portugal wildfires
2022	100%	Exceptional pan-Iberian heat and drought
2024	94%	Near-record Douro growing season

In contrast, known cool and wet years (1999, 2002, 2008, 2014, 2019) register $\leq 3\%$ of sites stressed, confirming the label logic is clean and not noise.

The primary pain is not only crop loss—it is post-event *cash-flow pressure*. The critical window is the period between a damaging event and the arrival of any compensation or credit. Traditional crop insurance is too slow for this problem.

3.2 Target Customers and Market Size

Primary customers: Small and medium wine producers in the Douro Valley, particularly family farms and cooperative members. The region counts approximately 19,000–21,000 registered vine-growers, predominantly with holdings of 1–3 hectares ivv2023, ivdp2022. This group combines high climate exposure with limited financial reserves.

The risk model reveals a clear sub-regional exposure gradient:

Table 2: Climate stress probability by IVDP sub-region

Sub-region	Climate stress probability	Climate profile
Baixo Corgo	45%	Cooler, wetter ($\sim 1,045$ mm/year)
Cima Corgo	55%	Intermediate
Douro Superior	72%	Hottest and driest (~ 596 mm/year)

Douro Superior producers face the highest exposure but the least capacity to absorb a bad year—the most underserved segment.

3.3 Competitive Positioning

Table 3: Competitive alternatives and their weaknesses

Alternative	Key weakness
Self-insurance	Requires reserves small producers typically lack
Public disaster aid	Uncertain timing, often delayed by months
Bank credit	Increases debt exactly under the worst financial conditions
Traditional crop insurance	Field assessment: slow, expensive, not designed for smallholders
Cooperative emergency support	Limited funds, uneven availability, not systematic

3.4 Business Model (B2B2C)

VinhaGuard operates as a **hybrid SaaS + insurance distribution layer**, not a regulated insurer. The B2B2C structure is shown in Table 4.

Table 4: B2B2C role structure

Layer	Role
VinhaGuard	Risk model, premium engine, trigger monitoring, user dashboard
Insurance partner	Underwriting capacity, regulatory legitimacy, distribution
End customer	Policy holder; receives automatic payout on trigger breach

Revenue streams:

- **Commission on gross written premium:** approximately 12% of GWP, aligned with portfolio growth.
- **Annual platform fee per insured vineyard:** EUR 50/year for dashboard, data pipeline, and monitoring.

VinhaGuard does not carry underwriting risk in the MVP or at scale. The insurer partner holds the risk balance sheet. This is the distinction that makes the platform economics viable for a startup.

4 Data Pipeline and Sources

4.1 Overview

The data pipeline is fully implemented and reproducible from the project repository. It produces a canonical ML dataset of 960 location-year rows from 32 Douro vineyard locations over 30 years (1995–2024).


```

1 # Stage 1: Generate vineyard location registry
2 python -m src.data.locations          # -> data/locations.csv
3
4 # Stage 2: Fetch daily ERA5 weather per site via Open-Meteo
5 python -m src.data.fetch_weather     # -> data/raw/weather/{id}.
    parquet
6
7 # Stage 3: Engineer annual heat / frost / drought features
8 python -m src.data.build_features    # -> data/interim/features.
    parquet
9
10 # Stage 4: Enrich with TerraClimate water-balance variables
11 python -m src.data.fetch_terraclimate # -> data/processed/
    terraclimate.parquet
12
13 # Stage 5: Merge, label, and write final ML dataset
14 python -m src.data.make_dataset      # -> data/processed/
    vinhaguard_dataset.parquet

```

Listing 1: Data pipeline stages

4.2 Primary Data Sources

Open-Meteo Historical Weather API (ERA5)

The primary weather data provider. The API distributes ERA5 reanalysis at approximately 9 km spatial resolution via the endpoint `archive-api.open-meteo.com/v1/archive`. Variables fetched: daily maximum, minimum, and mean temperature; daily precipitation sum. Period: 1995-01-01 to 2024-12-31. Licensed for non-commercial use with attribution zippenfenig2023.

ERA5 is the fifth-generation ECMWF global atmospheric reanalysis, covering 1940 to the present at hourly resolution on a ~31 km native grid hersbach2020.

TerraClimate

Monthly global climate and climatic water-balance data at approximately 4 km resolution, provided by the Climatology Lab at UC Merced abatzoglou2018. VinhaGuard uses it to add physiologically meaningful drought indicators not captured by daily rainfall counts alone. Variables used: vapour pressure deficit (`vpd`), climate water deficit (`def`), soil moisture (`soil`), and precipitation (`ppt`). Data is read via OPeNDAP and cached locally.

IVDP Sub-region Definitions

The Instituto dos Vinhos do Douro e do Porto (IVDP) three-zone classification (Baixo Corgo, Cima Corgo, Douro Superior) is used as the primary geographic grouping variable throughout the dataset and product ivdp2022.

IPMA Climate Normals

The Instituto Português do Mar e da Atmosfera (IPMA) published climate normals and extreme event archives were used to cross-validate the labelling rule against independently documented stress years.

4.3 Attempted Data Request: IVV Historical Damage Records

As part of the data collection effort, the team contacted the Instituto da Vinha e do Vinho (IVV) at ivv@ivv.gov.pt, requesting access to historical regional damage or yield-loss records for the Douro Demarcated Region. Such records would have allowed us to validate the ML target label (currently a climate-trigger proxy) against real observed agricultural losses, substantially strengthening the actuarial credibility of the pricing model.

The IVV did not respond to this request within the project timeline. As a result, the team proceeded with the climate-trigger proxy label derived from ERA5 features as the ML target. This is a known limitation of the prototype: the model’s high AUC (0.977) reflects the recovery of the label rule from weather features, not predictive power over actual crop losses. Production deployment would require claims or yield-loss data from an insurer or agricultural authority partner.

4.4 Dataset Summary

Table 5: Canonical ML dataset characteristics

Attribute	Value
File	<code>data/processed/vinhaguard_dataset.parquet</code>
Rows	960 location-year observations
Columns	29 (20 weather/geography + 8 TerraClimate + 1 target)
Locations	32 Douro vineyard sites
Period	1995–2024 (30 years)
Target	<code>climate_stress_year</code> (binary)
Positive class share	39.8%

Feature groups in the dataset:

- **Geography:** sub-region, latitude, longitude, elevation (m).
- **Heat:** heat-day counts at 30, 35, 38 °C thresholds; summer temperature statistics; heatwave maximum streak.
- **Frost:** spring frost days; spring severe frost days ($\leq -2^{\circ}\text{C}$); minimum spring overnight temperature.
- **Drought:** annual and summer precipitation; dry-day counts; maximum consecutive dry days; growing-degree days.
- **TerraClimate water balance:** vapour pressure deficit, climate water deficit, soil moisture, and precipitation (growing season and summer aggregates).

4.5 Target Label Definition

The binary target `climate_stress_year` is set to 1 when *any one* of the following per-location anomaly conditions is met:

- heat days above 35 °C exceed the location’s own 80th-percentile threshold, *or*
- spring severe frost days ($\leq -2^{\circ}\text{C}$) exceed the location’s 80th-percentile threshold, *or*
- maximum consecutive dry days exceed the location’s 80th-percentile threshold.

The use of per-location percentile thresholds—rather than fixed absolute values—is the core design choice. It ensures triggers reflect relative anomaly for each specific vineyard’s climate baseline rather than a uniform Douro-wide standard.

4.6 Known Data Limitations

1. **Missing locations.** Weather data was retrieved for 32 of 36 planned sites. Four locations (CC03, CC12, DS05, DS10) could not be fetched within the project timeline due to Open-Meteo free-tier rate limits. Sub-region coverage remains balanced.
2. **ERA5 grid resolution.** The approximately 9 km grid means two vineyards within the same cell receive identical raw weather values, regardless of micro-climate differences. This is a source of basis risk.
3. **TerraClimate resolution.** Soil moisture and water deficit are estimated at approximately 4 km grid scale and do not capture plot-level terrace structure, irrigation, or canopy management.
4. **Proxy label.** The target is derived from the same ERA5 features used as model inputs. Production use requires claims or yield validation data (e.g., from IVV or an insurer partner).
5. **No NDVI.** Sentinel-2 vegetation stress imagery was scoped out of the MVP due to time constraints but is on the roadmap.

5 Machine Learning Methodology

5.1 Model Selection

Two models were trained in `model/train.py`:

- **Logistic Regression:** interpretable baseline with standardised features, one-hot sub-region encoding, and class-balanced weights.
- **Random Forest:** main deployed model; captures non-linear trigger interactions; class-balanced; shallow enough to remain interpretable via feature importance.

A third variant, the **calibrated Random Forest** (isotonic calibration), is the deployed artifact. Calibration is essential because the pricing backend uses model risk probability as a direct input to the expected-payout formula.

5.2 Evaluation Design

The evaluation uses two complementary strategies:

1. **Chronological holdout:** train on 1995–2019 (800 rows), test on 2020–2024 (160 rows). This is deliberately harder than a random split because the test set contains recent extreme years (2022 and 2024 are among the most stressed on record).
2. **GroupKFold cross-validation** by `location_id`: tests whether the model generalises across vineyard sites rather than memorising one location’s history.

Test-set class imbalance caveat. The 2020–2024 holdout has a positive class share of 71.9%, substantially above the full dataset’s 39.8%. This is because those five years were climatically extreme. Precision metrics are therefore optimistic estimates for an average year; the GroupKFold cross-validation provides the more stable estimate of generalisation performance.

5.3 Results

Table 6: Model performance on the chronological 2020–2024 holdout

Model	ROC-AUC	PR-AUC	Brier	Accuracy	Precision	Recall	F1
Logistic Regression	0.949	0.984	0.071	0.913	0.963	0.913	0.938
Random Forest	0.970	0.986	0.115	0.881	0.971	0.861	0.912
RF Calibrated	0.977	0.986	0.042	0.944	0.934	0.991	0.962

GroupKFold robustness check (Random Forest, by `location_id`): mean ROC-AUC = 0.937, standard deviation = 0.055 across five folds (fold scores: 0.816, 0.949, 0.967, 0.988, 0.956).

5.4 Feature Importance

Table 7 shows the top feature drivers by permutation importance:

Table 7: Top feature drivers (permutation importance, Random Forest)

Rank	Feature	Interpretation
1	<code>max_consecutive_dry_days</code>	Drought persistence is the dominant signal
2	<code>heatwave_max_streak</code>	Heat spell duration, not just day count
3	<code>elevation_m</code>	Elevation drives micro-climate differentiation
4	<code>heat_days_35</code>	Moderate heat accumulation
5	<code>spring_severe_frost_days</code>	Flowering-period frost exposure
6	<code>tc_def_growing_sum</code>	TerraClimate growing-season water deficit
7	<code>summer_precip_mm</code>	Summer rainfall total
8	<code>min_spring_tmin</code>	Coldest spring overnight temperature

Drought persistence (`max_consecutive_dry_days`) is the strongest driver—consistent with the region’s known sensitivity to multi-week dry spells. The appearance of TerraClimate’s growing-season water deficit (`tc_def_growing_sum`) confirms that the enrichment adds physiologically meaningful signal beyond simple day-count proxies.

5.5 Honest Caveat on the High AUC

Because `climate_stress_year` is a deterministic function of the same ERA5 features used as model inputs, the model partly recovers its own label rule from noisy aggregated inputs. The high ROC-AUC should be interpreted as confirming that the engineered features faithfully encode the trigger logic, not as evidence of out-of-sample predictive

power over actual crop losses. The known-year validation table (Section 3, Table 1) is the more meaningful real-world correctness check.

5.6 Risk Probability Logic at Quote Time

At quote time, next season’s weather is unknown. The pricing backend therefore does not fabricate future feature values. Instead, it:

1. filters historical location-year rows to the selected sub-region;
2. optionally weights rows by elevation similarity if an elevation is provided;
3. computes the relevant historical trigger frequency;
4. blends that rate with the model’s average stress probability on comparable historical rows:

$$\hat{p}(\text{trigger}) = 0.70 \times \text{historical trigger rate} + 0.30 \times \text{model stress probability}$$

This keeps the AI component real and explainable while avoiding fabricated future-weather assumptions.

6 Insurance Product Design

6.1 Parametric Trigger Definitions

Table 8: MVP parametric trigger conditions

Risk type	Trigger condition	Dataset feature
Heat	≥ 5 days above 38°C during growing season	heat_days_38, heatwave_1
Frost	≥ 3 days below -2°C during flowering (Mar–May)	spring_severe_frost_days
Drought	Max consecutive dry days $\geq$ site’s 80th-percentile threshold	max_consecutive_dry_days

6.2 Premium Formula

The annual premium follows the formula implemented in `pages/3_Pricing_Explainer.py`:

$$\text{Expected Loss} = \text{Insured Value} \times \hat{p}(\text{trigger}) \times \text{Loss Given Trigger}$$

$$\text{Risk Loading} = \text{Expected Loss} \times \text{Loading \%}$$

$$\text{Admin Cost} = 50 \text{ EUR} + 2 \text{ EUR} \times \text{Vineyard Area (ha)}$$

$$\text{Annual Premium} = (\text{Expected Loss} + \text{Risk Loading} + \text{Admin Cost}) \times (1 + \text{Admin Margin})$$

Default parameters by risk type:

Table 9: Pricing parameters by risk type

Risk type	Loss Given Trigger	Risk Loading	Admin Margin
Heat	50%	25%	15%
Frost	40%	25%	15%
Drought	45%	25%	15%
All (combined)	55%	30%	15%

6.3 Sub-regional Premium Examples

The following premiums are backend outputs from `predict_risk_and_premium()` for a EUR 5,000 payout policy on a 12-hectare vineyard. They are not illustrative assumptions:

Table 10: Annual premium by sub-region and risk type (EUR 5,000 payout, 12 ha)

Sub-region	Heat	Frost	Drought	All
Baixo Corgo	EUR 242 (4%)	EUR 169 (3%)	EUR 911 (25%)	EUR 1,553 (35%)
Cima Corgo	EUR 413 (9%)	EUR 323 (8%)	EUR 987 (28%)	EUR 1,817 (42%)
Douro Superior	EUR 1,006 (25%)	EUR 162 (3%)	EUR 957 (26%)	EUR 1,609 (36%)

Parentheses show the implied trigger probability used in pricing.

Heat-only coverage at EUR 242–413 for Baixo/Cima Corgo represents 5–8% of the EUR 5,000 payout value—affordable for a smallholder liquidity product. Douro Superior heat coverage at EUR 1,006 (20% of payout) is less affordable, and the pilot should test whether a lower payout amount (EUR 2,500) or a higher threshold (8+ days above 38 °C) could bring the premium to EUR 500–600.

6.4 Basis Risk

Basis risk—the probability that a trigger fires but no real loss occurs, or vice versa—is estimated dynamically in `model/predict.py`:

$$\text{Basis Risk} \approx 8\text{--}22\%$$

starting at a base of 10%, increasing when sub-region coverage is thinner, and increasing further when the entered elevation deviates substantially from the historical sub-region median. This dynamic estimation—rather than a fixed constant—is one of the three reasons the product requires ML rather than a static actuarial table (Section 1).

7 Unit Economics and Business Viability

7.1 Revenue and Cost Structure

Revenue: 12% commission on gross written premium plus EUR 50/year platform fee per insured vineyard.

Main cost drivers: ERA5/Open-Meteo API access; risk model retraining and monitoring; Streamlit dashboard hosting; compliance support and partner management.

7.2 Break-even Analysis

Two scenarios are modelled, using a conservative blended average premium of EUR 400 (reflecting a heat-only entry product mix):

Table 11: Scenario A — Lean project cost base (EUR 80,000 fixed costs)

Producers	GWP (EUR)	VG Revenue	Fixed Costs	EBIT
50 (pilot)	27,500	5,800	80,000	−74,200
200	110,000	23,200	83,000	−59,800
500	275,000	58,000	90,000	−32,000
833 (BE)	457,000	96,500	96,500	0
1,000	550,000	116,000	100,000	+16,000
2,000	1,100,000	232,000	120,000	+112,000

Table 12: Scenario B — Regulated insurance infrastructure (EUR 350,000 fixed costs)

Producers	VG Revenue	Fixed Costs	EBIT
50 (pilot)	5,800	351,000	−345,200
500	58,000	360,000	−302,000
1,000	116,000	370,000	−254,000
3,646 (BE)	420,000	420,000	0
5,000	575,000	450,000	+125,000

Scenario A break-even (~833 producers) is achievable as a lean student-project spinout. A commercially compliant insurance infrastructure product breaks even closer to 3,500–4,000 producers—a Year 3–5 target.

Important note: Both scenarios represent *platform break-even only*, not insurance profitability. The insurer partner’s profitability requires separate actuarial validation of trigger frequency, loss correlation, and reinsurance cost.

7.3 Go-to-Market Phasing

Table 13: Recommended go-to-market phasing

Phase	Focus	Target	Key milestone
Pilot (Year 1)	Heat trigger, Douro Superior	50–100 via 1 cooperative	Validate trigger trust, W payout experience
Expand (Year 2)	Frost + drought; Cima Corgo	200–500	Second insurer partner; proach BE
Scale (Year 3+)	Full Douro; white-label	800–1,000+	Cross break-even; expl Alentejo

8 Prototype

8.1 Architecture

The prototype is a Streamlit multi-page application backed by a scikit-learn pricing engine and a Gemini-powered AI assistant. The full stack is implemented in Python 3.12 with the following key dependencies:

- `streamlit==1.45.0` — web application framework
- `scikit-learn==1.5.2` — ML training, calibration, inference
- `pandas==2.2.3`, `pyarrow==19.0.1` — data layer
- `plotly==5.24.1` — interactive charts
- `google-genai==2.0.1` — Gemini 2.5 Flash Lite AI assistant
- `xarray==2025.4.0`, `netCDF4==1.7.2` — TerraClimate data extraction

8.2 Application Pages

Page 1 — Risk Assessment

The user selects a Douro sub-region (or familiar alias such as Pinhão, Régua, or Vila Nova de Foz Côa), enters vineyard area and insured value, and optionally provides elevation. The backend calls `predict_risk_and_premium()` and displays: trigger probability, annual premium, basis-risk indicator, and a plain-language explanation of the trigger condition in Portuguese.

Page 2 — Climate Dashboard

Interactive historical charts showing 30 years of trigger behaviour, the Random Forest’s model-performance diagnostics (ROC curve, precision–recall curve, calibration curve, confusion matrix), and feature-importance visualisation. The data source is `model/artifacts/scored_history.csv` —the same 960-row scored dataset used to train and evaluate the model.

Page 3 — Pricing Explainer

A transparent waterfall chart decomposing the premium into expected payout, risk loading, admin cost, and margin. Users can adjust key parameters interactively to understand the sensitivity of the price to assumptions.

Page 4 — AI Assistant (Chatbot)

A Gemini 2.5 Flash Lite powered assistant constrained by a strict system prompt to answer only questions about: VinhaGuard product functionality; parametric insurance concepts; Douro climate-risk context; and app usage guidance. It cannot modify premiums, triggers, or policy terms, and it clearly states that VinhaGuard is an academic prototype rather than a licensed insurance product.

8.3 Running the Prototype

```
1 python -m venv .venv
2 source .venv/bin/activate
3 pip install -r requirements.txt
```



```
4 streamlit run app.py
```

Listing 2: Running the Streamlit app

To retrain the model from the committed processed data:

```
1 python -m model.train
```

Listing 3: Retraining the risk model

9 AI Safety, Transparency, and Governance

Table 14: Risk register and safeguards

Risk	Why it matters	Safeguard
Hallucinated policy explanation	Producer misunderstands coverage	Retrieval-grounded assistant; policy terms fixed in contract; no binding LLM promises
Basis risk	Real loss may not trigger payout	Explicit pre-purchase disclosure; dynamic basis-risk indicator in app
Proxy label	Model trained on derived labels, not real losses	Honest documentation; known-year validation; IVV data request made
Overreliance on AI premium	Insurer accepts model output without review	Human underwriting approval required before policy issuance
Correlated payout exposure	Extreme years trigger mass simultaneous payouts	VinhaGuard carries no underwriting risk; insurer/reinsurer manages correlated exposure
Regulatory misclassification	Platform mistaken for unlicensed insurer	Explicitly positioned as infrastructure layer; insurer partner holds regulated role

All premium recommendations display the full formula and inputs. The app logs model inputs and pricing outputs for auditability. A clear disclaimer is shown throughout: policy terms are set at purchase; the LLM assistant is for explanation only and cannot modify policy conditions.

10 GenAI Transparency Log

The project used generative AI tools throughout. All use is disclosed here, as required by the project brief. The full log is also committed to the repository at docs/genai_log.md.

10.1 Tools Used

Table 15: GenAI tools and their roles in the project

Tool	Used for
Claude / Claude Code	Early project structuring; business drafting; data-pipeline scaffolding; frontend scaffolding; code generation, debugging, and documentation
ChatGPT / Codex	Repository review; TerraClimate research and integration; ML training workflow; pricing backend review; documentation cleanup; final consistency p
Gemini 2.5 Flash Lite	In-app AI assistant for product/support questions in the Streamlit prototype

10.2 Business and Insurance Work

GenAI assisted with: synthesising the business problem and target customer; drafting and revising the business report; checking that the premium formula was consistent across the report and app; developing the unit-economics notebook structure; clarifying the distinction between climate-stress probability and payout trigger probability; and documenting basis risk, go-to-market logic, and regulatory limitations.

Human decisions included: positioning VinhaGuard as a B2B2C infrastructure layer rather than a regulated insurer; selecting parametric insurance as the product format; defining the MVP as fast climate-liquidity protection; and choosing the revenue model and pilot assumptions.

10.3 Ideation Work

During the ideation phase, Claude and ChatGPT were used to generate a broad list of AI-native startup concepts and to stress-test them against feasibility and data availability criteria. The GovBid AI and VendorBot concepts described in Section 2 emerged from this process. The team reviewed the AI-generated assessments and made the final decision to pursue VinhaGuard.

10.4 Data Pipeline and ML Work

GenAI assisted with: repository structure recommendations; scripts for location generation, weather fetching, feature engineering, and target labelling; API debugging for Open-Meteo rate limits; implementing `model/train.py`; adding TerraClimate water-balance features; replacing mock pricing logic with a trained-model backend; and generating evaluation charts.

Human-reviewed decisions included: the choice of chronological train/test split; GroupK-Fold by location as a robustness check; deployment of the calibrated Random Forest; use of per-location percentile thresholds; and honest framing of the target as a climate-trigger proxy.

10.5 Frontend Work

GenAI assisted with: Streamlit page scaffolding; UI copy and layout; integration with `model.predict.predict_risk_and_premium()`; pricing explainer charts; and chatbot system-prompt drafting.

Human decisions included: the final page structure and product workflow; the decision to show three canonical IVDP risk profiles; the wording of key disclaimers and basis-risk explanations; and the guardrail design for the chatbot.

10.6 Important Corrections Made During Review

- Replaced early mock risk outputs with the trained model backend.
- Added TerraClimate water-balance data so drought risk was not based only on rainfall-count assumptions.
- Aligned heat-trigger wording around the severe-event threshold of at least 5 days above 38 °C.
- Removed unprofessional role-numbered references from public documentation.
- Updated README and documentation to match the final deployable architecture.

10.7 Limitations of GenAI Assistance

GenAI-generated drafts required careful human review. The most important risks encountered were: overconfident wording around model accuracy; confusion between climate-stress probability and payout-trigger probability; stale documentation after model and data changes; and possible hallucinated implementation details if documentation was not checked against code. All identified issues were corrected before submission.

11 Team Contributions and Internal Point Allocation

The project was developed collaboratively by all group members. The team operated with a flat contribution structure: each member actively participated in ideation, data pipeline development, model training, frontend integration, business analysis, and documentation.

In accordance with the project requirements for team-based evaluation, the team declares that **points are divided evenly among all members, as each member contributed equally to the project**. The total number of points allocated sums to $100 \times N_{\text{members}}$ as required, with each member assigned 100 points.

Table 16: Internal point allocation

Member	Points assigned	Notes
Member 1	100	Equal contribution
Member 2	100	Equal contribution
Member 3	100	Equal contribution <i>Adjust member count and total to match</i>
Member 4	100	Equal contribution
Member 5	100	Equal contribution
Total	500	$100 \times 5 = 500$ <i>your actual group size.</i>

12 Conclusion

VinhaGuard demonstrates that a substantive, deployable ML product can be built on publicly available climate data within an academic project timeline. The system addresses

a real market gap: the absence of fast, transparent, affordable parametric climate insurance for small Douro Valley wine producers.

The core ML contribution is not a chatbot wrapper but a domain-specific climate feature pipeline with per-location percentile trigger calibration, a calibrated Random Forest risk model, and a transparent premium pricing backend. The prototype achieves strong classification performance (ROC-AUC 0.977 calibrated) and correctly identifies all five documented extreme vintages.

The primary honest limitation is that the ML target is a climate-trigger proxy derived from the same ERA5 features used as inputs, not real claims or yield-loss data. An attempt was made to obtain historical damage records from IVV (ivv@ivv.gov.pt) but received no response. Production deployment would require claims data from an insurer or agricultural authority partner—precisely the kind of partnership the B2B2C model is designed to enable.

The moat is the per-location percentile calibration logic embedded in the current codebase, which becomes more valuable as operating seasons accumulate proprietary trigger-versus-loss records. The insurer workflow integration and cooperative distribution network are roadmap items that create durable switching costs once built.

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